```python
import numpy as np
from statsmodels.stats.proportion import proportions_ztest
import matplotlib.pyplot as plt
from scipy.stats import norm

# Given data
sales = np.array([120, 150])      # Number of conversions
users = np.array([1000, 1200])    # Total users
alpha = 0.05

# Conversion rates
rate_A = sales[0] / users[0]
rate_B = sales[1] / users[1]

# Two-proportion Z-test
z_stat, p_value = proportions_ztest(sales, users)

# Critical z-value (two-tailed)
z_critical = norm.ppf(1 - alpha/2)

# Display results
print("A/B Testing Using Two-Proportion Z-Test")
print("-" * 45)
print(f"Version A Users           : {users[0]}")
print(f"Version A Sales           : {sales[0]}")
print(f"Version A Conversion Rate : {rate_A:.4f} ({rate_A*100:.2f}%)")

print(f"\nVersion B Users           : {users[1]}")
print(f"Version B Sales           : {sales[1]}")
print(f"Version B Conversion Rate : {rate_B:.4f} ({rate_B*100:.2f}%)")

print(f"\nZ-statistic              : {z_stat:.4f}")
print(f"P-value                  : {p_value:.4f}")
print(f"Critical Z-value         : ±{z_critical:.4f}")
```

```
A/B Testing Using Two-Proportion Z-Test
---------------------------------------------
Version A Users           : 1000
Version A Sales           : 120
Version A Conversion Rate : 0.1200 (12.00%)

Version B Users           : 1200
Version B Sales           : 150
Version B Conversion Rate : 0.1250 (12.50%)

Z-statistic              : -0.3559
P-value                  : 0.7219
Critical Z-value         : ±1.9600
```

```python
# Decision
print("\nDecision:")
```

```
if p_value < alpha:
    print("Reject the null hypothesis (H0).")
    print("Conclusion: There is a statistically significant difference
between the conversion rates.")
else:
    print("Fail to reject the null hypothesis (H0).")
    print("Conclusion: There is no statistically significant
difference between the conversion rates.")
```

```
Decision:
Fail to reject the null hypothesis (H0).
Conclusion: There is no statistically significant difference between
the conversion rates.
```

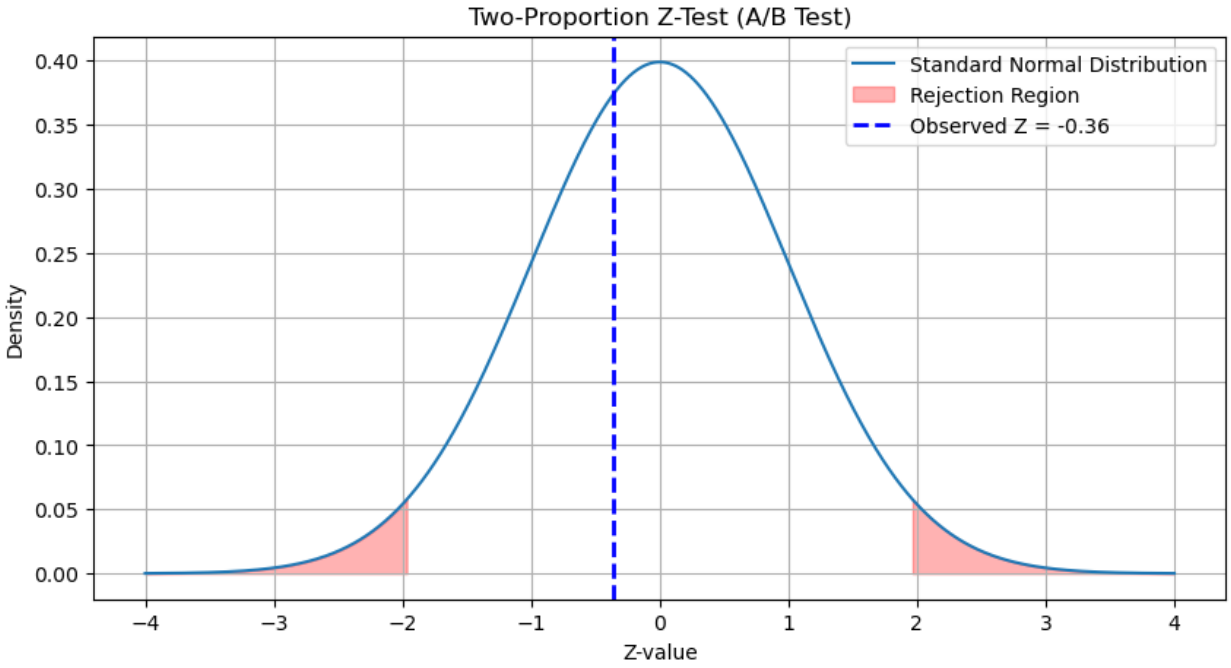
```
# Plot the standard normal distribution
x = np.linspace(-4, 4, 1000)
y = norm.pdf(x)

plt.figure(figsize=(10,5))
plt.plot(x, y, label='Standard Normal Distribution')

# Shade rejection regions
plt.fill_between(x, 0, y, where=(x <= -z_critical),
                 color='red', alpha=0.3, label='Rejection Region')
plt.fill_between(x, 0, y, where=(x >= z_critical),
                 color='red', alpha=0.3)

# Observed Z statistic
plt.axvline(z_stat, color='blue', linestyle='--',
            linewidth=2, label=f'Observed Z = {z_stat:.2f}')

plt.title("Two-Proportion Z-Test (A/B Test)")
plt.xlabel("Z-value")
plt.ylabel("Density")
plt.grid(True)
plt.legend()
plt.show()
```


Two-Proportion Z-Test (A/B Test)
Standard Normal Distribution
Rejection Region
Observed Z = -0.36
Density
Z-value
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
−4
−3
−2
−1
0
1
2
3
4